

**International Islamic University Chittagong**  
**Center for General Education (CGED)**  
Midterm Examination, Autumn- 2023  
**Course Code:** URED- 3503 (URED- 3101 for LLB)  
**Course Title:** Political Thoughts and Social Behavior

Time: 1.5 Hours

Full Marks: 30

[Answer **any three** of the following. All questions are of equal value]

Q.1. Define Islamic political system and discuss the main principles of it with proper evidences from the *Qur'an* and *Sunnah*.

Q.2. Define *shariah* and then analyze the sources of *shariah* with some examples.

Or

Define Islamic constitution and describe the special features of it with some details.

Q.3. What is the qur'anic and sunnetic term for the chief executive? Describe his qualifications with proper arguments.

Q.4. How many kinds of *majlish ash-shura* in an Islamic state? Describe the functions of the members of *shura* in details.

# International Islamic University Chittagong

## Morality Development Program (MDP)

Mid Term Examination, Autumn-2023

Semester: 5th

Course Code: **MDP-3505**    Course Title: **Concept on Moral Development - II**

Time: 1.5 hours

Full Marks: 30

[Answer all the following questions]

1. Describe in brief how do students have the potential to play significant roles in the formation of an ideal society. 10
  2. Discuss about the negative effects of dowry system in the society. Explain the ways of removing the dowry system in the society. 10
  - 3.(a) What are the misconceptions about Islam? Which factors are playing a role in creating misconceptions about Islam? Discuss about them. 5
  - (b) Discuss about the ways of removing the misconceptions about Islam. 5
- OR
- c). Signify the recreation and entertainment in the light of Quran and Sunnah. 5
  - d). Describe the concept of Halal entrepreneurship with reference from Islamic Shariah. 5

**International Islamic University Chittagong**  
**Department of Computer Science and Engineering**  
B.Sc in CSE                      Mid Examination, Autumn 2023  
**Course Code: CSE-3523**

**Course Title: Microprocessors, Microcontrollers and Embedded Systems**

**Time: 1 hour 30 minutes**

**Total Marks: 30**

- 
- Answer all **Three** of the following questions.
  - The figures in the right-hand margin indicate full marks.

1 a) What are the essential components of a microprocessor? What makes the microprocessor different from the microcontroller? 3 CLO 1 U

b) Suppose you need to run a high-quality and speedy performance video on computers. What type of bus you will use to support this operation on the computer? Describe the bus. 3 CLO 1 AP

c) You are sitting at a specific place inside a bus. The ticket checker asks you to show your ticket. 4 CLO 1 AP  
Do you see any similarity between address, control, and data bus with the above-mentioned scenario? If yes, explain with logic.

2 a) In the microprocessor, what is the function of IP and ALU? 2 CLO 1 R

or, What determines whether a microprocessor is considered an 8-bit, 16-bit, or 32-bit device?

b) What is memory segmentation for the 8086 processor? What is the use of it? How can a 16-bit microprocessor generate 20-bit memory addresses? 4 CLO 1 R

or, Draw the Internal Block Diagram of the 8086 Microprocessor.

c) If the stack segment register contains 56C0H and the stack pointer register contains 1358H, what is the physical address of the top of the stack? 2 CLO 1 AP  
*57958H*

d) The value of the Code Segment (CS) Register is 4042H and the value of different offsets is as follows: 2

BX: 2025H, IP: 0580H, DI: 4247H

Calculate the effective address of the memory location pointed by the CS register.  
*40740H*

3 a) Differentiate between Mnemonic and Pseudo-op.

1 CLO 1 U

4 CLO 1 E

b) Consider the following instructions:

- A. MOV AX, BL
- B. MOV BL, 0032H
- C. MOV AX, [BX]
- D. MOV AX, CS

What are the addressing modes used in the above statements? Are the above instructions correct or incorrect? Explain if the instructions are right or wrong with the logic.

or, Which of the following are legal numbers or legal names? If the number or name is illegal explain the reason.

I. Numbers:

- A. 1,101B
- B. FFFFH
- C. -55D
- D. 1FH

II. Names:

- A. iandhe
- B. i&he
- C. A45.26
- D. A2

c) Write an assembly language program in Microprocessor 8086 to compute:

5 CLO 2 AP

- I.  $Y = A + B$
- II.  $X = 2Y - 3Z$

**International Islamic University Chittagong**  
**Department of Computer Science and Engineering**  
*B. Sc. in CSE Mid-term Examination, Autumn 2023*  
**Course Code: CSE 3529 Title: System Analysis and Design**  
**Total marks: 30 Time: 1 hour 30 minutes**

[Answer all the questions. Figures in the right-hand margin indicate full marks.]

| Course Outcomes (COs) of the Questions |                                                                  |
|----------------------------------------|------------------------------------------------------------------|
| CO1                                    | Understand project management and system development life cycle. |
| CO2                                    | Analyze the requirement and feasibility of software application  |
| CO3                                    | Design an information system.                                    |
| CO4                                    | Compile the system design document                               |

| Bloom's Levels of the Questions |          |            |       |          |          |        |
|---------------------------------|----------|------------|-------|----------|----------|--------|
| Letter Symbols                  | R        | U          | Ap    | An       | E        | C      |
| Meaning                         | Remember | Understand | Apply | Analysis | Evaluate | Create |

- 1.a) "Foodpanda" is a popular food delivery app that allows users to order food from various restaurants. Considering the following statements aligned with the app, classify them into *Strategic*, *Tactical*, and *Operational* type of information. Justify your answer. 5 CO2 DL U

- i. In-app pop-up notifications (immediately when the user opens the app) highlighting the new features
- ii. Added in-app features of "top 10 most ordered meals of the week", and "top rated restaurants near you"
- iii. Showing trendiest feature in app, i.e. calorie count for every item

- b) Write characteristics of quality information. 5 CO3 C
- Suppose an online learning platform seeks to enhance its user engagement and course completion rates. So, a system analyst designs lucrative features such as progress tracking, interactive quizzes, and discussion forums to promote student involvement. What do you think what kind of roles/responsibilities can a system analyst play in such cases according to the above scenario?

- 2.a) Which SDLC model would you adopt for developing an e-commerce platform that requires small and there are no uncertain requirements? Justify your answer reasoning with the advantages and disadvantages of your model. 5 C01 U

Or,

Suppose you want to develop an online learning platform. It develops modules on various subjects, enabling instructors to create and deliver content, and students to engage in interactive learning. Regular feedback from educators helps refine the platform's features and usability. Which SDLC model will be more suitable for the above learning platform? Justify your answer

- b) Distinguish between Methodologies, Models, Tools and Techniques with example. 5 C01 U

- 3.a) What is the significance of feasibility analysis in project management? Which phase of project management requires feasibility analysis? Briefly discuss about three major areas to conduct a feasibility study of an organization. 5 C01 U

- b) Describe the steps to conduct the economic feasibility analysis with proper example. 5 C02 An

Or

Define the following:

- i) Deliverables
- ii) Break-Even Point (with mathematical explanation)
- iii) Net Present Value (with mathematical explanation)
- iv) Recurring cost
- v) Return on Investment (with mathematical explanation)

*Bismillahir Rahmanir Rahim*

# International Islamic University Chittagong

Department of Computer Science and Engineering

Mid-Term Examination, Autumn-2023

Course: **CSE-3521 (Computer Architecture)**

Time: **1 hour and 30 Minutes**

Marks: **30**

[Answer all *three* questions; Figures in the right hand margin indicate Marks]

- Q1. a) Define Computer Architecture. Explain the subcategories of computer architecture 1+2 CO1
- b) A program runs in 10 seconds on computer A, which has a 300 MHz clock. We are trying to help a computer designer build a machine, B, that will run this program in 6 seconds. The designer has determined that a substantial increase in the clock rate is possible, but this increase will affect the rest of the CPU design, causing machine B to require 1.2 times as many clock cycles as machine A for this program. What clock rate should we tell the designer to target? 4 CO2
- c) Suppose we have two implementations of the same instruction set architecture. Computer1 has a CPI of 2.0 and a clock cycle time of 250 ps for some programs, and computer 2 has a CPI of 1.2 and a clock cycle time of 500 ps for same programs. Which computer is faster for this program and by how much? 3 CO2
- Q2. a) State four design principles of MIPS 2 CO1
- b) Translate the following MIPS Assembly Language instructions into Machine L 4 CO2
- (i) sub \$s2, \$t5, 20
  - (ii) bne \$s3, \$s1, 100
  - (iii) sw \$s7, 80(\$s2)
  - (iv) slt \$s4, \$s6, \$s7

c) What is the difference between CISC and RISC computers? 2 CO1  
Or  
Which registers are reserved for assembler and operating system?

d) Describe the MIPS addressing mode with figure. 2 CO1

Q3. a) Describe the first version of division Algorithm using Flow Chart. 3 CO1

Or

Describe the Final version of Multiplication Algorithm using Flow Chart.

b) Show the IEEE 754 binary representation of the number  $-0.25X_{ten}$  in single and double precision 3 CO2

c) Divide  $5_{ten} \div 3_{ten}$  using final version of ~~multiplication~~ <sup>division</sup> algorithm 4 CO2

$$\begin{array}{r} 1 \\ 3 \overline{) 5} \\ \underline{3} \phantom{0} \\ 2 \phantom{0} \end{array}$$

# International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering (CSE)

Mid-term Examination, Autumn-2023

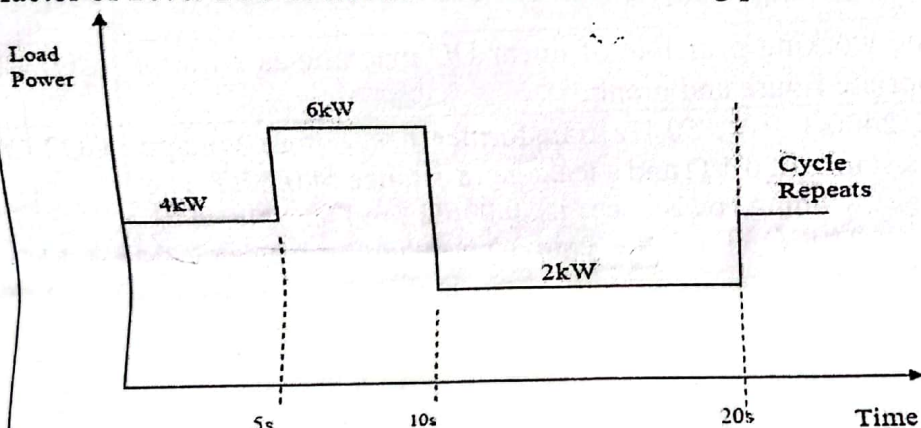
Program: BSc in CSE

Course Title: Electrical Drives and Instrumentation

Course Code: EEE-2421

Time: 1 hour 30 minutes

Full Marks: 30

|      |                                                                                                                                                                                                                                                                                                                                                                                                              |     |    |   |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|---|
| 1(a) | State the faraday's second law. Describe different types of motor duty class with appropriate timing diagram.                                                                                                                                                                                                                                                                                                | CO1 | U  | 3 |
| 1(b) | <p>What do you mean by motor sizing? The load torque profile shown in figure 1(b) was measured for a constant speed load. Choose an appropriately sized motor to drive this load assuming a service factor of 20%. You do not need to consider starting performance.</p>  <p style="text-align: center;">Figure: 1(b)</p> | CO2 | An | 3 |
| 1(c) | Describe the operation of DC generator with proper net sketch.                                                                                                                                                                                                                                                                                                                                               | CO1 | R  | 4 |

|      |                                                                                                                                                                                                                                                                                                                                                                                                      |     |    |   |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|---|
| 2(a) | Draw the equivalent circuit of transformer with proper notation.                                                                                                                                                                                                                                                                                                                                     | CO1 | U  | 3 |
| 2(b) | When a transformer is connected to a 1000-V, 50-Hz supply the core loss is 1000 W, of which 650 is hysteresis and 350 is eddy current loss. If the applied voltage is raised to 2000 V and the frequency to 100 Hz, find the new core losses.                                                                                                                                                        | CO2 | An | 3 |
| 2(c) | <p>In no-load test of single phase transformer, the following test data were obtained: Primary voltage: 220V; Secondary voltage: 110V; Primary Current: 0.5A; Power Input: 30W. Resistance of the primary winding = 0.6 ohm</p> <p>Find the following:</p> <p>i) The turn ratio ii) The magnetizing component of no load current<br/>iii) Its working (or loss) component<br/>iv) The iron loss.</p> | CO2 | An | 4 |

OR

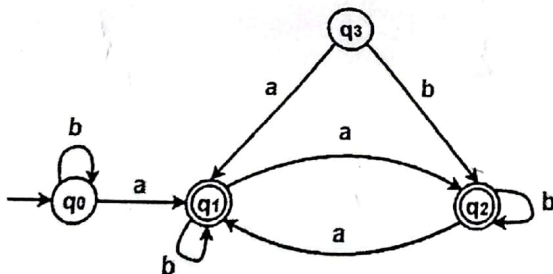
|      |                                                                                                                                                                                                                                                      |     |    |   |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|---|
| 2(a) | Define transformer. Discuss the working principle of a transformer.                                                                                                                                                                                  | CO1 | R  | 3 |
| 2(b) | Why transformer test is needed? Compare the characteristics of an Ideal transformer and a practical transformer.                                                                                                                                     | CO1 | U  | 3 |
| 2(c) | A 5kVA 200/1000 V, 50Hz, single-phase transformer gave the following test result:<br>O.C. Test (L.V. Side): 200V, 1.2A, 90W<br>S.C. Test (H.V. Side): 50V, 5A, 110W<br>Calculate the parameters of the equivalent circuit referred to the L.V. side. | CO2 | An | 4 |

|      |                                                                                                                                                                                                                                                                                                                                                                            |     |    |   |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|---|
| 3(a) | Describe the hysteresis losses and eddy current loss for transformer.                                                                                                                                                                                                                                                                                                      | CO1 | U  | 2 |
| 3(b) | Describe the working principle of linear DC machine as a Motor with appropriate figure and graph.                                                                                                                                                                                                                                                                          | CO1 | R  | 4 |
| 3(c) | A 30kVA, 2400/120-V, 50-Hz transformer has a high voltage winding resistance of $0.1 \Omega$ and a leakage resistance of $0.22 \Omega$ . The low voltage winding resistance is $0.035 \Omega$ and the leakage resistance is $0.012 \Omega$ . Find the equivalent winding resistance, reactance and impedance referred to the i) high voltage side and ii) low voltage side | CO2 | An | 4 |



**International Islamic University Chittagong**  
Department of Computer Science and Engineering  
B. Sc. in CSE Midterm Examination, Autumn 2023  
Course Code: CSE-3527 Course Title: Compiler  
Total marks: 30 Time: 90 minutes

1. a. What is bytecode? How do Java language processors work? 2 CO1 C2  
b. How is an executable target program generated? Explain it using a language processing system. 3 CO1 C2  
c. Translate the following assignment statement step by step using compiler phases: 4 CO1 C3  
 $z = a * b + c / 4;$  [z, a, b, and c contain floating point values]
2. a. Consider the CFG:  $S \rightarrow a + SS \mid * SS \mid S \mid a$  2 CO1 C1  
1. Show that:  $aa^+a^*$  can be generated by the grammar.  $+ a^* aa$  C2  
2. Draw the parse tree  
b. Consider the grammar:  $P \rightarrow aPbP \mid bPaP \mid \epsilon$  4 CO1 C3  
Consider **ababab** as the string to check the ambiguity of the given grammar.  
c. Construct the regular expressions and provide two accepted strings for each. 4 CO2 C3  
(Any two):  
1)  $\{w \mid w: \text{starts with a 'b' and has at least one a}\}$   
2)  $\{w \mid w: \text{does not contain the substring abab}\}$   
3)  $\{w \mid w: \text{set of strings consisting of even number of a's followed by b}\}$   
4)  $\{w \mid w: \text{set of strings of 1's and 0's in even length}\}$
3. a. Minimize the following DFA: 4 CO2 C3



- b. Convert the following regular expression directly to its equivalent NFA: 2 CO2 C3  
 $abb(a|b)^*ba$   
OR,  
 $(a|b)^*a(a|b)(a|b)$

- c. Convert the following NFA to DFA: 4 CO1 C3

